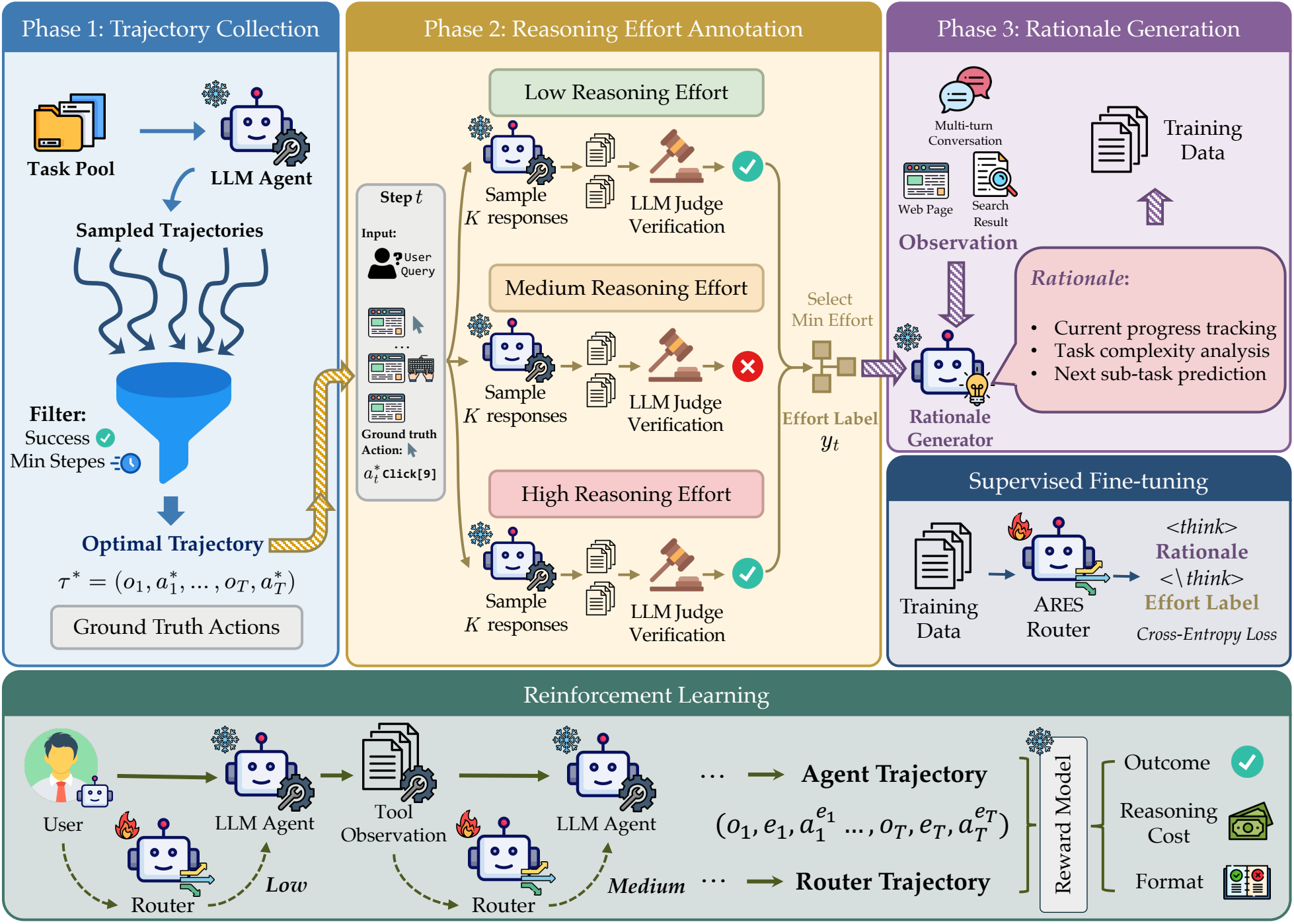

Phase 1: Trajectory Collection
Task Pool
LLM Agent
Sampled Trajectories
Filter:
Success
Min Stepes
Optimal Trajectory
$\tau^* = (o_1, a_1^*, \ldots, o_T, a_T^*)$
Ground Truth Actions
Phase 2: Reasoning Effort Annotation
Step t
Input:
User Query
Ground truth Action:
a_t^* Click[9]
Low Reasoning Effort
Medium Reasoning Effort
High Reasoning Effort
Sample K responses
LLM Judge Verification
Select Min Effort
Effort Label y_t
Phase 3: Rationale Generation
Multi-turn Conversation
Web Page
Search Result
Observation
Training Data
Rationale:
Current progress tracking
Task complexity analysis
Next sub-task prediction
Rationale Generator
Supervised Fine-tuning
Training Data
ARES Router
<think>
Rationale
<\think>
Effort Label
Cross-Entropy Loss
Reinforcement Learning
User
LLM Agent
Router
Low
Tool Observation
LLM Agent
Router
Medium
Agent Trajectory
$(o_1, e_1, a_1^{e_1} \ldots, o_T, e_T, a_T^{e_T})$
Router Trajectory
Reward Model
Outcome
Reasoning Cost
Format